

LIFEMOTE BROADCOM-BASED CPE AGENT FIRMWARE SPECIFICATION

Version	Date	Author	Reviewed by
v2.0	Dec 24 2019	Eren Soyak	
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1 Scope

This document contains the specification for the requirements of the Lifemote data gathering agent on Broadcom-based CPEs. The requirements are grouped as those relating to Deployment and Commands. All requirements are to be considered mandatory, and no part shall remain unfulfilled without losing Lifemote functionality.

2 Deployment

The Lifemote data gathering agent shall be downloaded by the CPE via HTTP each time the CPE boots, and if for whatever reason the agent is terminated during runtime. The software module implementing this functionality is called the Lifemote Deployer, and is available for download from Lifemote.

2.1 TR-069 Options

Lifemote agent shall be toggled on and off via a TR-069 option. The TR-069 options supported by the CPE shall be as follows:

- **Device.X.LIFEMOTE_EXT.LifemoteAgent.Enable:** Boolean, Read/Write. When set to True, will immediately activate the Lifemote Deployer specified in Sec. 2.2, and also add the Lifemote Deployer launch to the boot script of the CPE, such that the Lifemote Deployer will be launched as a background process each time the CPE boots. When set to False, will immediately terminate the Lifemote Deployer background process using the SIGTERM signal¹, and remove the Lifemote Deployer launch from the boot script of the CPE such that the Lifemote Deployer will not be launched the next time the CPE reboots. Note that toggling this option from True to False and back to True should prompt an update of the Lifemote agent.
- **Device.X.LIFEMOTE_EXT.LifemoteAgent.URL:** String, Read/Write. Specifies the URL from which the Lifemote agent will be downloaded by the Lifemote Deployer. This string shall be recorded as an environment variable named `LIFEMOTE_AGENT_URL` in a manner accessible by the Lifemote Deployer. When this field is set, the Lifemote Deployer shall be automatically terminated using the SIGTERM signal and restarted. This behavior shall allow a new Lifemote Agent to be deployed via a single TR-069 command.

¹Note that the background process shall be terminated with SIGTERM and not SIGKILL so that the Lifemote Deployer can run cleanup routines upon intercepting the SIGTERM signal.

- `Device.X.LIFEMOTE_EXT.LifemoteAgent.State`: Boolean, Read only. The CPE shall set this parameter to True if the Lifemote Deployer background process is running, and to False otherwise.

2.2 Lifemote Deployer

The Lifemote Deployer is a small software module that shall be run as a background process. This module periodically checks for a running Lifemote agent on the CPE, and when it finds the agent to be absent will download the agent via HTTP GET of the `LIFEMOTE_AGENT_URL` provided via TR-069 and run it as a background process.

If the Lifemote Deployer cannot download the agent via HTTP GET, it shall keep retrying at increasing intervals. The first interval shall be uniformly distributed between 5-10 minutes, then 10-20 minutes, then 20-40 minutes, then 40-80 minutes, and finally 60-120 minutes. The retries shall remain at every 60-120 minutes after this point until the agent is successfully downloaded. The randomization for this functionality is meant to spread out the loading of the server offering the agent.

If `LIFEMOTE_AGENT_URL` is empty or undefined, the Lifemote Deployer shall terminate immediately upon launch. This behavior shall be visible via the `Device.X.LIFEMOTE_EXT.LifemoteAgent.State` TR-069 parameter. Note that `LIFEMOTE_AGENT_URL` can be modified simply by the `Device.X.LIFEMOTE_EXT.LifemoteAgent.URL` without setting `Device.X.LIFEMOTE_EXT.LifemoteAgent.Enable`.

Sample BASH code for the Lifemote Deployer can be downloaded from Lifemote². The CPE vendor is encouraged to use this code directly, although any code with exact equivalent functionality is acceptable.

Note: As an additional security measure the Lifemote Deployer shall terminate itself if it receives an HTTP return code of 403, signifying it is attempting to download an agent from an URL for which it was not whitelisted.

3 Encryption

The Lifemote agent shall be downloaded by the Lifemote Deployer via HTTP, and send its updates via HTTP GET or POST queries encrypted by TLS.

4 Commands

The Lifemote agent shall gather data via calling the Broadcom driver interfaces (`wl` for WiFi and `xds1ctl` for DSL where necessary), Linux executables such as `brctl` and `arp`, and reading Linux system files such as the DHCP leases table. The output text shall be parsed and encoded into JSON format, and sent to the Lifemote cloud via HTTP POST.

The Lifemote agent shall exercise the following commands during its operation – functionality of these shall be tested by the CPE vendor.

1. `curl` (capable of HTTP GET and POST and `-k -s -g -m` extensions)
2. `echo`, `cat`, `tr`, `grep`, `egrep`, `cut`, `head`, `tail`, `date`, `sleep`, `ps`, `kill`, `read`, `hexdump`
3. `ifconfig`, `route`
4. `dnsmasq`, `udhcp` or equivalent mechanism to view DHCP leases

²<http://dream.demo.lifemote.com/v1/scripts?type=phoenix>

5. a mechanism to read remaining lease time of WAN IP when WAN IP is obtained by the CPE via DHCP client
6. brctl show,
brctl showmacs,
brctl showstp
7. arp
8. openssl v1.0.1 or newer
9. xdsctl
10. wl commands:
wl assoclist
wl sta_info (which provides the tx total bytes and rx data bytes fields)
wl rssi
wl counters
wl chanim_stats
wl noise
wl scan / wl scanresults
wl status